

Name: _____

MA 2320: Quiz 1 Prep
Calculus I Review

1. Compute the following antiderivatives:

(a) $\int 2x^2 + 1 \, dx$

(b) $\int 3\sqrt[3]{x} - \sin(x) \, dx$

(c) $\int \frac{3x-x^2+1}{2x} \, dx$

(d) $\int e^x \, dx$

(e) $\int e^{3x} \, dx$

(f) $\int 3x\sqrt[3]{x^2-1} \, dx$

(g) $\int \frac{\sin(x)}{1+\sin^2(x)} \, dx$

(h) $\int \frac{\sin(x)}{1-\sin^2(x)} \, dx$ No u-sub needed here

(i) $\int \frac{3\sin(x)}{4-5\cos(x)} \, dx$

(j) $\int 5x\sqrt{2x-9} \, dx$

(k) $\int \sec(x) \tan(x) \sqrt{\sec(x)+1} \, dx$